

Intramyocellular lipid concentrations are correlated with insulin sensitivity in humans: a ^1H NMR spectroscopy study.

Recent muscle biopsy studies have shown a relation between intramuscular lipid content and insulin resistance. The aim of this study was to test this relation in humans by using a novel proton nuclear magnetic resonance (^1H NMR) spectroscopy technique, which enables non-invasive and rapid (approximately 45 min) determination of intramyocellular lipid (IMCL) content. Normal weight non-diabetic adults ($n = 23$, age 29 ± 2 years, BMI = 24.1 ± 0.5 kg/m²) were studied using cross-sectional analysis. Insulin sensitivity was assessed by a 2-h hyperinsulinaemic (approximately 450 pmol/l)-euglycaemic (approximately 5 mmol/l) clamp test. Intramyocellular lipid concentrations were determined by using localized ^1H NMR spectroscopy of soleus muscle. Simple linear regression analysis showed an inverse correlation ($r = -0.579$, $p = 0.0037$) [corrected] between intramyocellular lipid content and M-value (100-120 min of clamp) as well as between fasting plasma non-esterified fatty acid concentration and M-value ($r = -0.54$, $p = 0.0267$). Intramyocellular lipid content was not related to BMI, age and fasting plasma concentrations of triglycerides, non-esterified fatty acids, glucose or insulin. These results show that intramyocellular lipid concentration, as assessed non invasively by localized ^1H NMR spectroscopy, is a good indicator of whole body insulin sensitivity in non-diabetic, non-obese humans.